

HEQU, China

WMO: 535640

Lat: 39.3649N

Lon: 111.2130E

Elev: 3402

StdP: 12.98

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.4	-1.7	-18.2	2.3	6.2	-14.2	2.9	8.6	16.2	19.8	13.6	26.0	1.0	20	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	92.7	65.9	90.0	65.3	87.7	64.7	73.1	83.4	71.6	81.5	70.1	79.7	5.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.6	127.9	76.9	68.9	120.7	75.7	67.2	113.8	75.1	39.4	83.5	37.8	81.4	36.5	79.7	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
16.4	13.8	11.5	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-10.7	97.1	5.1	3.1	-14.4	99.3	-17.4	101.2	-20.2	102.9	-24.0	105.2
			-11.2	76.1	5.0	1.7	-14.8	77.2	-17.7	78.2	-20.5	79.2	-24.1	80.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.7	15.5	25.4	39.3	53.7	64.1	72.2	75.5	71.8	62.1	49.3	34.1	19.7		
	DBStd	21.63	7.44	8.99	8.42	8.29	6.72	5.01	4.06	4.58	5.95	6.97	7.48	7.33		
	HDD50	3674	1071	688	343	56	1	0	0	0	2	97	477	939		
	HDD65	6836	1536	1108	796	349	97	7	0	5	122	486	926	1404		
	CDD50	3189	0	0	13	166	438	665	791	676	364	76	0	0		
	CDD65	877	0	0	0	9	69	222	326	216	34	1	0	0		
	CDH74	9261	0	0	3	231	1110	2497	3160	1818	422	20	0	0		
	CDH80	3540	0	0	0	59	371	1042	1363	607	98	0	0	0		
	WSAvg	3.7	2.4	3.2	4.6	5.4	5.2	4.4	3.5	3.1	3.2	3.3	3.2	2.7		
	PrecAvg	15.7	0.1	0.2	0.4	0.7	1.3	2.0	4.1	3.4	2.1	0.9	0.4	0.1		
Precipitation	PrecMax	25.6	0.4	0.5	2.3	2.1	2.8	3.9	12.0	7.4	4.3	3.4	2.1	0.5		
	PrecMin	9.3	0.0	0.0	0.0	0.0	0.1	0.2	1.3	0.4	0.5	0.1	0.0	0.0		
	PrecStd	4.4	0.1	0.2	0.5	0.6	0.7	0.9	2.4	1.9	1.1	0.7	0.5	0.1		
	PrecStd	4.4	0.1	0.2	0.5	0.6	0.7	0.9	2.4	1.9	1.1	0.7	0.5	0.1		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.6	55.6	72.1	87.1	91.9	95.9	96.4	92.4	88.3	76.8	61.3	42.6		
		MCWB	31.2	39.5	48.7	56.3	59.6	63.4	69.7	69.1	62.7	58.1	44.5	31.7		
	2%	DB	36.8	49.9	65.3	80.2	87.9	92.2	93.2	89.2	82.4	72.2	55.9	38.5		
		MCWB	28.0	36.5	44.7	53.0	58.3	62.7	68.3	68.5	61.9	54.0	42.6	29.9		
	5%	DB	32.9	45.5	60.4	75.6	84.3	89.2	90.4	86.2	78.7	67.8	51.7	35.4		
		MCWB	25.5	33.8	42.1	51.3	56.7	62.3	68.0	68.2	61.0	52.2	40.7	28.1		
	10%	DB	29.3	41.2	55.7	70.9	80.6	86.3	87.7	83.4	75.3	63.6	47.7	32.3		
		MCWB	23.2	31.5	39.8	49.1	55.2	61.7	68.0	67.6	59.9	50.6	38.8	26.5		
	0.4%	WB	32.1	41.0	50.0	59.1	63.6	70.3	75.6	74.7	69.6	59.8	48.5	33.7		
		MCDB	40.2	52.8	68.3	75.6	81.2	82.9	85.2	84.0	77.4	72.6	53.9	39.2		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.7	37.9	46.0	56.2	61.2	68.1	73.8	72.8	66.4	56.7	45.1	31.1		
		MCDB	36.1	47.8	61.5	72.0	77.8	79.9	84.3	82.3	75.1	66.8	52.4	36.7		
	5%	WB	25.9	34.9	43.3	53.7	59.4	66.5	72.1	71.4	64.2	54.6	42.4	28.9		
		MCDB	32.5	43.6	57.8	70.6	76.1	79.1	82.8	80.1	73.0	64.2	49.5	34.8		
	10%	WB	23.3	32.1	40.7	51.4	57.8	65.0	70.8	69.9	62.2	52.3	39.6	26.6		
		MCDB	29.2	40.4	53.4	67.1	74.5	78.1	81.3	78.1	71.8	60.7	47.0	32.4		
Mean Daily Temperature Range	5% DB	MDBR	24.0	25.6	25.4	26.6	26.3	24.5	21.8	21.4	23.6	24.3	21.9	21.7		
		MCDBR	27.1	30.0	32.3	33.9	33.2	31.0	27.5	26.4	29.8	30.5	27.5	24.0		
		MCWBR	19.7	18.4	16.5	14.3	11.7	9.5	8.7	9.4	11.9	14.6	15.6	16.4		
	5% WB	MCDBR	25.9	27.2	29.7	27.1	25.6	22.5	20.7	20.2	21.3	24.6	22.7	22.2		
		MCWBR	19.1	17.1	15.9	12.2	10.2	8.6	8.7	8.9	10.0	12.6	13.8	15.6		
Clear-Sky Solar Irradiance	taub		0.346	0.415	0.443	0.487	0.517	0.581	0.561	0.564	0.515	0.461	0.377	0.348		
	taud		2.114	1.963	1.917	1.870	1.852	1.763	1.869	1.840	1.894	1.951	2.103	2.145		
	Ebn at Noon		251	246	254	252	247	231	234	228	230	227	236	239		
	Edh at Noon		36	48	56	62	64	70	63	63	56	48	36	32		
All-Sky Solar Radiation	RadAvg		865	1128	1560	1880	2090	1962	1860	1685	1426	1199	911	751		
	RadStd		68	81	130	122	129	152	86	111	99	90	113	79		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	-158	-167	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	-2.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page